This assignment will not be graded and is only for practice.

Level 1

1) Let L be an affine subspace of R^3 defined as $L = \{(x, y, z) \mid x + y + 2z = 6\}$, then which of the following subspaces of R^3 corresponds to the affine subspace L? ***1 point***

- $\{(x, y, z) \mid x + y + z = 5\}$
- $\{(x, y, z) \mid x + y + 2z = 0\}$
- $\{(x, y, z) \mid x + 2y + z = 0\}$
- $\{(x, y, z) \mid 2x + y + z = 0\}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\{(x, y, z) \mid x + y + 2z = 0\}$

2) Which of the following subsets of R^2 represent an affine subspace of R^2? ***1 point***

- $\{(x, y) \mid x^2 + y^2 = 0\}$
- $\{(x, y) \mid x^2 + y^2 = 4\}$
- $\{(x, y) \mid x^5 + 1 = y\}$
- $\{(x, y) \mid 3x + y = 1\}$
- $\{(x, y) \mid x = 1\}$
- $\{(x, y) \mid y = 5x^2\}$
- $\{(x, y) \mid y = x^3 + 2\}$
- $\{(1, 2)\}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\{(x, y) \mid x^2 + y^2 = 0\}$
$\{(x, y) \mid 3x + y = 1\}$
$\{(x, y) \mid x = 1\}$
$\{(1, 2)\}$

3) Which of following subsets of $\mathbb{R}^3$ represent an affine subspace of $\mathbb{R}^3$? **1 point**

- [ ] $\{(x, y, z) \mid x^2 + y^2 + z^2 = 0\}$
- [ ] $\{(x, y, z) \mid x^2 + z^2 + y = 1\}$
- [ ] $\{(x, y, z) \mid x + y + z = 2\}$
- [ ] $\{(x, y, z) \mid 2x + y = 3z\}$
- [ ] $\{(x, y, z) \mid xy = z\}$
- [ ] $\{(x, y, z) \mid y = x + z\}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\{(x, y, z) \mid x^2 + y^2 + z^2 = 0\}$
$\{(x, y, z) \mid x + y + z = 2\}$
$\{(x, y, z) \mid 2x + y = 3z\}$
$\{(x, y, z) \mid y = x + z\}$

4) Let U be a subspace of the vector space $\mathbb{R}^3$ and a basis of U is given by $\{(0, 1, -3), (1, 0, -1)\}$. Then which of the following subsets of $\mathbb{R}^3$ is an affine subspace of $\mathbb{R}^3$ such that the corresponding vector subspace is U? **1 point**

- [ ] $L = \{(x, y, z) \mid x + 3y + z = 3\}$
- [ ] $L = \{(x, y, z) \mid x + 3y + 2z = 0\}$
- [ ] $L = \{(x, y, z) \mid x + 3y + z = 0\}$
- [ ] $L = \{(x, y, z) \mid x + 3y + z = 8\}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$L = \{(x, y, z) \mid x + 3y + z = 3\}$
$L = \{(x, y, z) \mid x + 3y + z = 0\}$
$L = \{(x, y, z) \mid x + 3y + z = 8\}$

5) Which of the following mappings is an affine mapping from $\mathbb{R}^2$ to an affine subspace of $\mathbb{R}^3$? ***1 point***

- [ ] $T(x, y) = (x + 1, y, xy)$
- [ ] $T(x, y) = (x, y, x + y + 5)$
- [ ] $T(x, y) = (2x - y + 2, 3y + 3, x + y - 1)$
- [ ] $T(x, y) = (3x, y^2, x)$
- [ ] $T(x, y) = (x, x + y, 0)$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$T(x, y) = (x, y, x + y + 5)$
$T(x, y) = (2x - y + 2, 3y + 3, x + y - 1)$
$T(x, y) = (x, x + y, 0)$

Level 2

6) Consider a system of linear equations ***1 point***

$$\begin{aligned} -x + y - z &= 1 \\ x - y + z &= -1 \\ x + z &= 0 \end{aligned}$$

Which of the following is an affine subspace L of $\mathbb{R}^3$ such that if $v \in L$, then v is a solution of the system of linear equations?

- $\{(t, -1, t) \mid t \in \mathbb{R}\}$
- $\{(-t, 1, t) \mid t \in \mathbb{R}\}$
- $\{(-t, 1, -t) \mid t \in \mathbb{R}\}$
- $\{(-t, -1, t) \mid t \in \mathbb{R}\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(-t, 1, t) \mid t \in \mathbb{R}\}$

7) Consider the following affine mappings from $\mathbb{R}^3$ to an affine subspace of $\mathbb{R}^3$ ***1 point***

$f_1(x, y, z) = (x + 1, x + 2y - 2, x + y + 2z)$
$f_2(x, y, z) = (x + 5, x + 2y - 3, x + y + 2z + 8)$
$f_3(x, y, z) = (2x + 4, y + z - 1, x + y + 5)$.

Which of the following option(s) is(are) true?

- [ ] A linear transformation corresponding to f_1 is $T(x, y, z) = (x, x + 2y, x + y + 2z)$
- [ ] The linear transformations corresponding to f_1 and f_2 are not the same.
- [ ] A linear transformation corresponding to f_3 is $T(x, y, z) = (x, y + z, x + y)$
- [ ] A linear transformation corresponding to f_3 is $T(x, y, z) = (2x, y + z, x + y)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

A linear transformation corresponding to f_1 is $T(x, y, z) = (x, x + 2y, x + y + 2z)$
A linear transformation corresponding to f_3 is $T(x, y, z) = (2x, y + z, x + y)$

8) Suppose $f : \mathbb{R}^2 \to \mathbb{R}^2$ is an affine mapping such that $f(2, 3) = (3, -1)$,$f(2, 1) = (1, 2)$ and $f(1, 0) = (0, 1)$. If $f(-2, 5) = (a, b)$, then value of $a + b$ is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -9

1 point

Your score is: 0/8